AI/Automation Spend in Banking Sector Analysis

Research Paper

Data Analytics Project | Python | Pandas | NumPy | MySQL | SQL | Matplotlib | Seaborn | Power BI

Abstract

Artificial Intelligence and automation are becoming important technologies in the banking sector. Banks are investing in technologies such as Generative AI, Machine Learning, RPA + AI and AI Automation to improve operational efficiency, automate repetitive activities, reduce costs and support business processes.

This project focuses on analyzing AI and automation spending across banks, countries, regions, technologies, AI use cases and maturity levels. The dataset contains information about AI spending, estimated cost savings, ROI, automation rate, monthly FTE hours saved, AI projects and maturity.

The data is cleaned and analyzed using Python. Pandas is used for data processing, grouping and aggregation, while NumPy is used for numerical calculations and feature engineering. MySQL and SQL are used for structured database analysis. Matplotlib and Seaborn are used to create analytical visualizations and correlation analysis. Power BI can be used as the interactive dashboard layer for business reporting.

The cleaned project dataset contains 72 records covering 2020 to 2025, 12 countries, 4 regions, 27 banks, 10 AI use cases and 4 technologies. Total recorded AI spending is USD 395.31 million and estimated cost savings are USD 403.37 million. The overall average ROI is 3.03x and average automation rate is 48.27%.

Keywords: Artificial Intelligence, Banking, Automation, AI Investment, AI Spend, Cost Savings, ROI, Automation Rate, Data Analytics, Python, MySQL, SQL, Power BI

1. Introduction

Artificial Intelligence has become an important part of modern banking. Banks use AI and automation in areas such as fraud detection, AML monitoring, KYC automation, credit risk and underwriting, customer service, document processing, IT operations, code generation and marketing personalization.

Investment in AI is useful only when it produces measurable business value. Therefore, analyzing spending together with cost savings, ROI, automation rate and productivity provides a more complete view of AI adoption.

The project AI/Automation Spend in Banking Sector Analysis is developed to study these indicators across different years, regions, countries, banks, technologies, use cases and maturity levels. Data analytics helps convert the raw records into meaningful comparisons and business insights.

2. Problem Statement

Banking AI investment data contains multiple dimensions such as year, region, country, bank, AI use case, technology, spending, cost savings, ROI, automation rate, FTE hours saved, number of projects and maturity. Reviewing these details manually is difficult and time-consuming.

There is a need for a structured data analytics solution that can clean the raw data, calculate business metrics, compare AI investments and present the findings in an easy-to-understand format. The project addresses this requirement using Python, SQL, visualization tools and dashboard-based reporting.

3. Objectives

1. To analyze AI and automation spending in the banking sector from 2020 to 2025.
2. To compare AI spending across regions and countries.
3. To analyze bank-wise AI investment.

4. To compare AI spending across different technologies.
5. To analyze AI spending and project adoption across different use cases.
6. To evaluate estimated cost savings and net benefit.
7. To calculate and compare ROI and automation rates.
8. To analyze monthly FTE hours saved through automation.
9. To compare AI maturity categories such as Pilot, Scaling and Production.
10. To create derived business metrics such as net benefit, savings-to-spend ratio and total impact score.
11. To store and analyze the data using MySQL and SQL queries.
12. To create analytical visualizations and support an interactive Power BI dashboard.

4. Dataset Description

The project uses a CSV dataset containing banking AI and automation investment records. The raw and cleaned datasets contain 72 records. The cleaned dataset contains 16 columns after feature engineering.

Attribute	Project Details
Records	72
Years	2020–2025
Countries	12
Regions	4
Banks	27
AI Use Cases	10
Technologies	4
Maturity Categories	3

Main Dataset Columns

Column	Description
Year	Year of the AI/automation investment record
Region	Geographical region of the bank
Country	Country associated with the bank
Bank	Bank or banking organization
AI_Use_Case	Business or operational AI application
Technology	AI/automation technology used
AI_Spend_USD_Million	AI investment amount in USD million
Estimated_Cost_Savings_USD_Million	Estimated cost savings in USD million
ROI_Multiple	Return on investment multiple
Automation_Rate_Percent	Percentage of process automation
FTE_Hours_Saved_Monthly	Monthly employee hours saved
AI_Projects	Number of AI projects
Maturity	AI adoption maturity level

5. Technologies Used

Python

Python is used as the main programming language for loading the dataset, cleaning records, performing calculations, grouping data and generating business insights.

Pandas

Pandas is used for DataFrame operations, missing-value checks, duplicate detection, grouping, aggregation, descriptive statistics and exporting the cleaned dataset.

NumPy

NumPy is used for numerical calculations and feature engineering, including the savings-to-spend ratio and other derived metrics.

Matplotlib and Seaborn

Matplotlib and Seaborn are used to create line charts, bar charts, distribution plots, scatter plots and correlation heatmaps.

MySQL and SQL

MySQL is used to store the cleaned banking AI data in a structured database. SQL queries are used for totals, averages, country-wise, region-wise, year-wise, bank-wise, technology-wise and maturity-wise analysis.

Power BI

Power BI can be used to convert the analyzed data into an interactive management dashboard. KPI cards, charts and slicers allow users to explore spending, savings, ROI, automation and project trends.

6. Data Cleaning and Preprocessing

Data cleaning was performed before analysis to improve data quality and consistency. The notebook checks the dataset shape, data types, missing values and duplicate records.

- Checked the number of rows and columns.
- Checked missing values in all columns.
- Checked duplicate records.
- Standardized column names by removing extra spaces and special characters.
- Removed duplicate rows where present.
- Trimmed leading and trailing spaces from text fields.
- Converted numeric columns to appropriate numeric data types.
- Validated the dataset after cleaning.

The final cleaned dataset contains 72 records. In the supplied cleaned dataset, all 16 columns contain zero missing values and there are zero duplicate rows.

7. Feature Engineering

Feature engineering was used to create additional business indicators that are not directly available in the raw dataset.

Feature	Formula / Meaning
Net Benefit (USD Million)	Estimated Cost Savings − AI Spend

Savings-to-Spend Ratio	Estimated Cost Savings ÷ AI Spend
Total Impact Score	ROI × (1 + Automation Rate/100) × (1 + AI Projects/10)

These derived metrics provide additional ways to evaluate whether AI investment is producing financial and operational impact.

8. Exploratory Data Analysis

Exploratory Data Analysis was performed using year, region, country, bank, AI use case, technology, maturity, AI spending, estimated savings, ROI, automation rate, FTE hours saved and AI project count.

- Distribution of AI spending
- Year-wise AI spending
- Year-wise estimated cost savings
- Region-wise AI spending
- Country-wise AI spending
- Technology-wise AI spending
- AI projects by use case
- Average ROI by technology
- Average automation rate by maturity
- Top banks by AI spending
- Correlation between investment and impact metrics

9. Year-wise AI Spending Analysis

Year	AI Spend (USD M)	Savings (USD M)	Avg ROI	Avg Automation (%)
2020	25.00	24.52	2.87	52.33
2021	31.90	35.16	3.40	50.31
2022	58.76	62.20	3.01	43.28
2023	64.16	50.81	2.54	45.85
2024	91.98	101.27	3.14	50.77
2025	123.51	129.41	3.24	47.07

The dataset shows a clear increase in total AI spending over the study period. Spending rises from USD 25.00 million in 2020 to USD 123.51 million in 2025, the highest value in the supplied dataset. Estimated cost savings also reach their highest value in 2025 at USD 129.41 million.

10. Region-wise Analysis

Region	Records	AI Spend (USD M)	Savings (USD M)	Avg ROI
Europe	24	163.90	168.84	3.05
Asia-Pacific	24	112.90	107.81	2.96
North America	12	93.11	97.77	2.96

Middle East	12	25.40	28.95	3.21

Europe has the highest recorded AI spending in the project dataset at USD 163.90 million, followed by Asia-Pacific at USD 112.90 million. These values describe the records included in this project and should not be interpreted as official regional banking investment statistics.

11. Country-wise Analysis

Country	Records	AI Spend (USD M)	Savings (USD M)	Avg ROI
Canada	6	46.57	39.72	2.82
USA	6	46.54	58.05	3.10
France	6	41.64	63.79	3.66
Germany	6	41.22	26.90	2.69
Switzerland	6	41.11	34.47	2.68
UK	6	39.93	43.68	3.18
India	6	30.35	24.94	2.40
Japan	6	29.18	23.30	2.79
Australia	6	27.60	22.43	2.81
Singapore	6	25.77	37.14	3.86
Saudi Arabia	6	13.43	16.02	3.30
UAE	6	11.97	12.93	3.13

Canada records the highest total AI spending in the supplied dataset at USD 46.57 million, closely followed by the USA at USD 46.54 million. France has an average ROI of 3.66x, while Singapore records the highest country-level average ROI at 3.86x among the project records.

12. Bank-wise AI Spending Analysis

Bank	Records	AI Spend (USD M)	Savings (USD M)	Avg ROI
Bank of America	4	33.39	43.44	3.48
Credit Suisse	4	28.11	26.51	2.94
Societe Generale	4	27.43	43.48	3.58
TD Bank	3	26.14	15.55	2.80
Deutsche Bank	3	24.58	18.13	3.00
SMBC	4	22.38	18.72	3.05
HSBC	3	21.41	23.85	3.29
RBC	3	20.43	24.17	2.85
Westpac	3	17.26	14.92	2.99
Commerzbank	3	16.64	8.77	2.38

Bank of America has the highest recorded AI spending among the banks in this dataset, with USD 33.39 million. Credit Suisse and Societe Generale follow with USD 28.11 million and USD 27.43 million respectively.

13. AI Use Case Analysis

AI Use Case	Records	Spend (USD M)	Savings (USD M)	Projects	Avg ROI
Credit Risk & Underwriting	9	58.86	52.80	65	2.95
Back-office Workflow	8	54.57	43.05	91	2.62
IT Operations	9	45.26	48.71	76	2.83
Code Generation	7	45.11	31.92	75	2.93
Document Processing	9	43.90	56.53	73	3.61
AML Monitoring	6	39.46	38.45	62	2.53
Marketing Personalization	7	37.09	44.07	64	3.26
Customer Service Chatbot	8	32.20	45.85	76	3.36
KYC Automation	4	23.90	26.72	56	3.02
Fraud Detection	5	14.96	15.27	37	3.11

Credit Risk & Underwriting has the highest recorded AI spending at USD 58.86 million. Back-office Workflow has the highest combined number of AI projects at 91. Document Processing shows a strong average ROI of 3.61x in the supplied records.

14. Technology-wise Analysis

Technology	Records	Spend (USD M)	Savings (USD M)	Avg ROI	Avg Automation (%)
RPA + AI	21	122.37	126.30	3.12	51.57
Generative AI	22	114.40	121.84	3.19	47.75
Machine Learning	20	113.28	106.52	2.86	43.76
AI Automation	9	45.26	48.71	2.83	51.88

RPA + AI has the highest total recorded spending at USD 122.37 million. Generative AI has the highest average ROI among the four technologies at 3.19x, while RPA + AI has an average automation rate of 51.57%.

15. AI Maturity Analysis

Maturity	Records	Spend (USD M)	Savings (USD M)	Avg ROI	Avg Automation (%)
Pilot	27	153.27	164.68	3.10	49.22
Production	15	71.70	76.02	3.43	53.67
Scaling	30	170.34	162.67	2.78	44.71

Production-stage records show the highest average automation rate at 53.67% and the highest average ROI at 3.43x. Scaling has the largest total spending in the dataset at USD 170.34 million.

16. Cost Savings, ROI and Productivity Analysis

Across all 72 records, total AI spending is USD 395.31 million and estimated cost savings are USD 403.37 million. The resulting aggregate net benefit is USD 8.06 million.

Metric	Value
Total AI Spend	USD 395.31 million
Estimated Cost Savings	USD 403.37 million
Aggregate Net Benefit	USD 8.06 million
Average ROI	3.03x
Average Automation Rate	48.27%
Total AI Projects	675
Total Monthly FTE Hours Saved	49118

17. MySQL and SQL Analysis

The project includes a MySQL database named ai_automation_banking and a table named ai_automation_spend. SQL is used to perform structured analysis directly at the database layer.

- Counting total records
- Calculating total AI spend
- Calculating total estimated cost savings
- Calculating average ROI
- Calculating average automation rate
- Calculating total AI projects
- Country-wise and region-wise spending
- Year-wise spending
- Bank-wise spending
- AI use case analysis
- Technology-wise spending and automation
- Maturity-wise project, ROI and automation analysis
- Finding highest-ROI projects

The SQL layer complements the Python analysis by allowing the same business questions to be answered directly from the structured database.

18. Python–MySQL Connectivity

The supplied project contains a Python database connection script using mysql.connector. The script connects to the ai_automation_banking database and checks the ai_automation_spend table. This creates a connection between the database layer and Python analysis workflow, allowing database results to be retrieved programmatically.

19. Data Visualization

The project notebook creates analytical charts using Matplotlib and Seaborn. The main visualizations include:

- Distribution of AI spending
- Year-wise AI spending trend
- Year-wise estimated cost savings
- Region-wise AI spending
- Country-wise AI spending
- Technology-wise AI spending
- AI projects by use case
- Average ROI by technology
- Average automation rate by maturity
- Top banks by AI spending
- Correlation heatmap of investment and impact metrics
- AI spending versus estimated cost savings scatter plot
- Automation rate versus monthly FTE hours saved scatter plot

20. Correlation and Relationship Analysis

The correlation analysis shows a correlation of 0.76 between AI spending and estimated cost savings in the supplied records. This indicates a strong positive association in this dataset, but correlation alone does not prove that higher spending causes higher savings.

ROI and estimated cost savings show a correlation of 0.54. Automation rate and FTE hours saved have a low correlation of 0.04 in this dataset, showing that the available records do not indicate a strong linear relationship between these two variables.

21. Power BI Dashboard

Power BI can be used as the reporting and interactive visualization layer for this project. The dashboard can present high-level KPIs and allow users to filter the analysis by year, region, country, bank, technology, AI use case and maturity.

Dashboard Component	Purpose
KPI Cards	Total AI Spend, Cost Savings, Net Benefit, Average ROI, Automation Rate and AI Projects
Year-wise Trend	Track changes in AI spending and savings over time
Region/Country Analysis	Compare geographical investment patterns
Technology Analysis	Compare spend, ROI and automation by technology

Use Case Analysis	Identify high-spend and high-project AI applications
Maturity Analysis	Compare Pilot, Scaling and Production performance
Slicers	Enable interactive filtering by key dimensions

An interactive dashboard makes it easier for managers and analysts to move from overall KPIs to detailed investment and operational analysis.

22. Results

- The cleaned dataset contains 72 records covering 2020 to 2025.
- The dataset contains 12 countries, 4 regions, 27 banks, 10 AI use cases and 4 technologies.
- Total recorded AI spending is approximately USD 395.31 million.
- Estimated total cost savings are approximately USD 403.37 million.
- Aggregate net benefit is approximately USD 8.06 million.
- Average ROI is 3.03x.
- Average automation rate is 48.27%.
- Total AI projects represented in the records are 675.
- Total monthly FTE hours saved across the records are 49118.
- 2025 has the highest total AI spending in the study period.
- Europe has the highest total recorded spending among the four regions.
- RPA + AI has the highest total spending among the technologies.
- Generative AI has the highest average ROI among the technologies.
- Production maturity has the highest average automation rate.

23. Discussion

The analysis indicates that AI investment in the supplied banking records increased substantially between 2020 and 2025. Spending growth is accompanied by increased estimated savings, although the relationship between investment and business impact differs across technologies, countries, banks and use cases.

The technology analysis shows that RPA + AI has the largest spending volume, while Generative AI has the highest average ROI. The maturity analysis shows stronger average automation and ROI for Production-stage records compared with Pilot and Scaling records. This suggests that maturity is an important dimension when evaluating AI performance, although the dataset is descriptive and does not establish causality.

The project demonstrates that combining financial indicators with operational indicators such as automation rate, FTE hours saved and number of AI projects provides a broader view of AI adoption in banking.

24. Limitations

- The dataset contains only 72 records and may not represent the entire global banking sector.
- The values represent the records included in the project and should not be treated as official industry statistics.
- Estimated cost savings may differ from actual realized savings.
- The analysis is mainly descriptive and does not establish causal relationships.
- External factors such as regulatory changes, macroeconomic conditions and bank-specific strategies are not included.
- The dataset covers 2020 to 2025 and may not represent the latest future investment patterns.
- Country and regional comparisons depend on the distribution of records in the supplied dataset.

25. Future Scope

- Add more banks, countries and years to increase coverage.
- Add actual financial outcomes and realized savings where available.
- Add investor/vendor and AI solution provider analysis.
- Add employee productivity and workforce impact analysis.
- Use machine learning to predict future AI spending or expected savings.
- Perform time-series forecasting for AI investment trends.
- Add real-time or automated data refresh.
- Connect Power BI directly to MySQL for automated reporting.
- Add scenario analysis for different AI investment levels.
- Develop a scoring model for prioritizing high-impact AI initiatives.

26. Conclusion

The AI/Automation Spend in Banking Sector Analysis project demonstrates a complete practical data analytics workflow for evaluating AI investment and business impact in banking. The workflow begins with a raw CSV dataset, followed by data validation, cleaning, feature engineering and exploratory analysis.

Python, Pandas and NumPy are used for data processing and calculations. MySQL and SQL provide structured database analysis, while Matplotlib and Seaborn support graphical and statistical visualization. Power BI can then be used to present the results through an interactive management dashboard.

The supplied cleaned dataset contains 72 records from 2020 to 2025, covering 12 countries, 4 regions, 27 banks, 10 AI use cases and 4 technologies. The records represent USD 395.31 million in AI spending and USD 403.37 million in estimated cost savings, with an average ROI of 3.03x and average automation rate of 48.27%.

Overall, the project shows how data analytics can transform banking AI investment records into measurable business insights and support data-driven decisions about AI adoption, automation, cost efficiency and operational impact.

27. Project Workflow Summary

Stage	Tools / Output
Data Collection	CSV banking AI/automation dataset
Data Cleaning	Python, Pandas
Feature Engineering	Pandas, NumPy
EDA	Pandas, NumPy
Database Analysis	MySQL, SQL
Python–Database Connectivity	mysql.connector
Visualization	Matplotlib, Seaborn
Dashboard Reporting	Power BI
Business Insights	Investment, savings, ROI, automation and productivity analysis